

MS5612 Assignment 1

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1. Two-State Option Pricing Model

Cilantro, Inc. stock: $S_u = \$80$, $S_d = \$60$, $S_0 = \$70$, $r_f = 6\%$, $T = 1$ year.

Risk-neutral probability of the up state:

$$p = \frac{(1 + r_f)S_0 - S_d}{S_u - S_d} = \frac{1.06(70) - 60}{80 - 60} = \frac{14.20}{20} = 0.71$$

(a) Exercise price = \$50

$$C_u = \max(80 - 50, 0) = 30, \quad C_d = \max(60 - 50, 0) = 10$$
$$C_0 = \frac{pC_u + (1 - p)C_d}{1 + r_f} = \frac{0.71(30) + 0.29(10)}{1.06} = \frac{24.20}{1.06} = \boxed{\$22.83}$$

Exercise price = \$55

$$C_u = \max(80 - 55, 0) = 25, \quad C_d = \max(60 - 55, 0) = 5$$
$$C_0 = \frac{0.71(25) + 0.29(5)}{1.06} = \frac{19.20}{1.06} = \boxed{\$18.11}$$

2. Put-Call Parity

$S_0 = \$73$, $K = \$70$, $C = \$5.27$, $T = 3/12 = 0.25$ yr, $r = 2.6\%$ (continuous compounding).

Put-call parity:

$$C + Ke^{-rT} = P + S_0 \implies P = C + Ke^{-rT} - S_0$$

$$Ke^{-rT} = 70e^{-0.026(0.25)} = 70e^{-0.0065} = 69.5466$$

$$P = 5.27 + 69.5466 - 73 = \boxed{\$1.82}$$

3. Black-Scholes

$S_0 = \$58$, $K = \$60$, $r = 2.7\%$ (continuous), $T = 4/12 = 0.3333$ yr, $\sigma = 47\%$.

$$d_1 = \frac{\ln(S_0/K) + \left(r + \frac{\sigma^2}{2}\right)T}{\sigma\sqrt{T}} = \frac{\ln(58/60) + (0.027 + 0.11045)(0.3333)}{0.47\sqrt{0.3333}}$$

$$d_1 = \frac{-0.03390 + 0.04582}{0.27135} = \frac{0.01192}{0.27135} = 0.0439$$

$$d_2 = d_1 - \sigma\sqrt{T} = 0.0439 - 0.27135 = -0.2274$$

From the standard normal table:

$$N(d_1) = N(0.0439) \approx 0.5175, \quad N(d_2) = N(-0.2274) \approx 0.4100$$

$$Ke^{-rT} = 60e^{-0.027(0.3333)} = 60e^{-0.009} = 59.4624$$

$$C = S_0N(d_1) - Ke^{-rT}N(d_2) = 58(0.5175) - 59.4624(0.4100) = \boxed{\$5.64}$$

Put-call parity gives the put price:

$$P = C - S_0 + Ke^{-rT} = 5.64 - 58 + 59.4624 = \boxed{\$7.10}$$

4. Risk-Neutral Valuation

$S_0 = \$84$, $u = 1.18$, $d = 0.82$, $K = \$80$, $T = 1$ yr, $r_f = 8\%$.

$$S_u = 84(1.18) = 99.12, \quad S_d = 84(0.82) = 68.88$$

$$C_u = \max(99.12 - 80, 0) = 19.12, \quad C_d = \max(68.88 - 80, 0) = 0$$

Risk-neutral probability:

$$p = \frac{(1 + r_f) - d}{u - d} = \frac{1.08 - 0.82}{1.18 - 0.82} = \frac{0.26}{0.36} = 0.7222$$

$$C_0 = \frac{pC_u + (1 - p)C_d}{1 + r_f} = \frac{0.7222(19.12) + 0}{1.08} = \boxed{\$12.79}$$

5. Real Options

The right to build is a call option on the office building, with the construction cost $K = \$60$ million acting as the exercise price.

$$S_0 = \$53.5\text{M}, \quad S_u = \$65.5\text{M}, \quad S_d = \$50.2\text{M}, \quad r_f = 4.8\%, \quad T = 1 \text{ yr}$$

$$C_u = \max(65.5 - 60, 0) = 5.5, \quad C_d = \max(50.2 - 60, 0) = 0$$

Risk-neutral probability:

$$p = \frac{(1 + r_f)S_0 - S_d}{S_u - S_d} = \frac{1.048(53.5) - 50.2}{65.5 - 50.2} = \frac{5.868}{15.3} = 0.3835$$

Value of the real option:

$$C_0 = \frac{pC_u + (1 - p)C_d}{1 + r_f} = \frac{0.3835(5.5) + 0}{1.048} = \boxed{\$2.01 \text{ million}}$$

Decision

The real option to build is worth approximately \$2.01 million, which exceeds the competitor's offer of \$1.8 million. **J&R Construction should reject the offer** and retain the right to build, since it is worth more than what is being offered for it.